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$$y(x) := \sqrt{x}$$

$$clz(x) := \begin{cases} 32 & \text{if } x = 0 \\ 32 - \operatorname{trunc}\left(\log_2\left(\operatorname{trunc}(x) + \frac{2^{32}-1}{2^{32}}\right) + \frac{2^{32}-1}{2^{32}}\right) & \text{else} \end{cases}$$

$$msb(x) := 32 - clz(x)$$

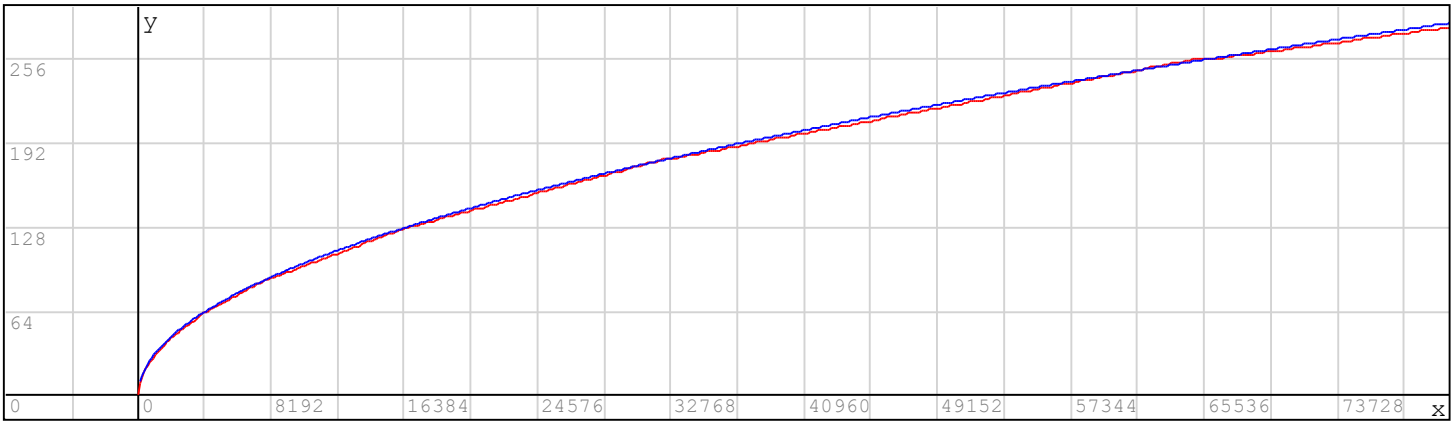
$$linScale(x) := \operatorname{trunc}\left(\frac{\operatorname{trunc}\left(\left(\sqrt{2}\right)^{msb(x)}\right) \cdot \operatorname{trunc}\left(\left(1+\sqrt{2}\right)\cdot 2^{29}\right)}{2^{32}}\right)$$

$$linSqrt(x) := \operatorname{trunc}\left(\frac{\left(\operatorname{trunc}(x) - 2^{msb(x)-1}\right)}{2^{msb(x)-1}} \cdot linScale(x)\right) + \operatorname{trunc}\left(\sqrt{2^{msb(x)-1}}\right)$$

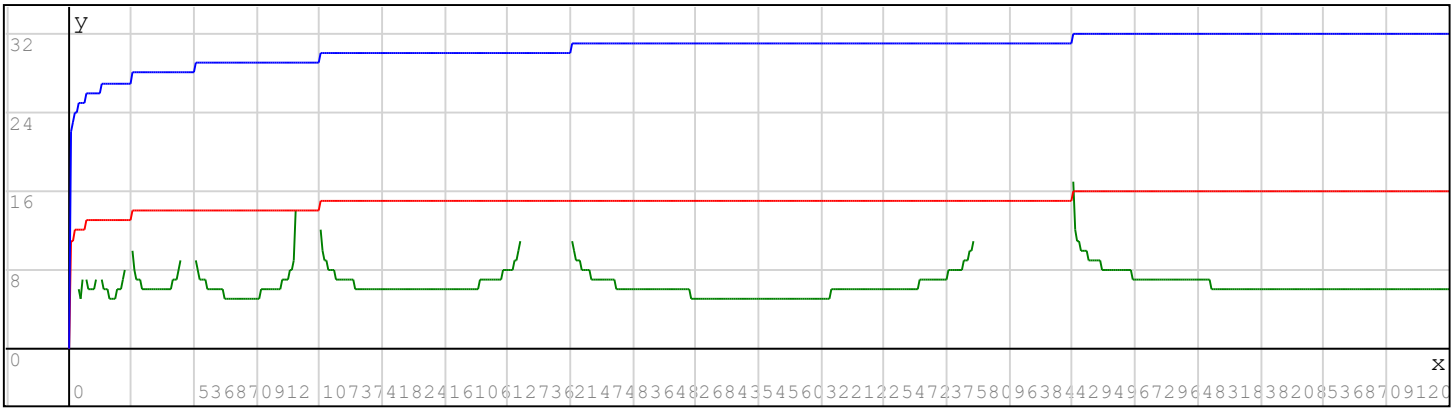
$$argBitCount(x) := \operatorname{trunc}\left(\log_2\left(x + \frac{2^{32}-1}{2^{32}}\right)\right)$$

$$resultBitCount(x) := \operatorname{trunc}\left(\log_2\left(\sqrt{\operatorname{trunc}(x)} + \frac{2^{32}-1}{2^{32}}\right)\right)$$

$$linearBitError(x) := \operatorname{trunc}\left(\log_2\left(\left(\sqrt{\operatorname{trunc}(x)} - linSqrt(x)\right)\right) + \frac{2^{32}-1}{2^{32}}\right)$$



$\sqrt{x}$
$linSqrt(x)$



$argBitCount(x)$
$resultBitCount(x)$
$resultBitCount(x) - linearBitError(x)$

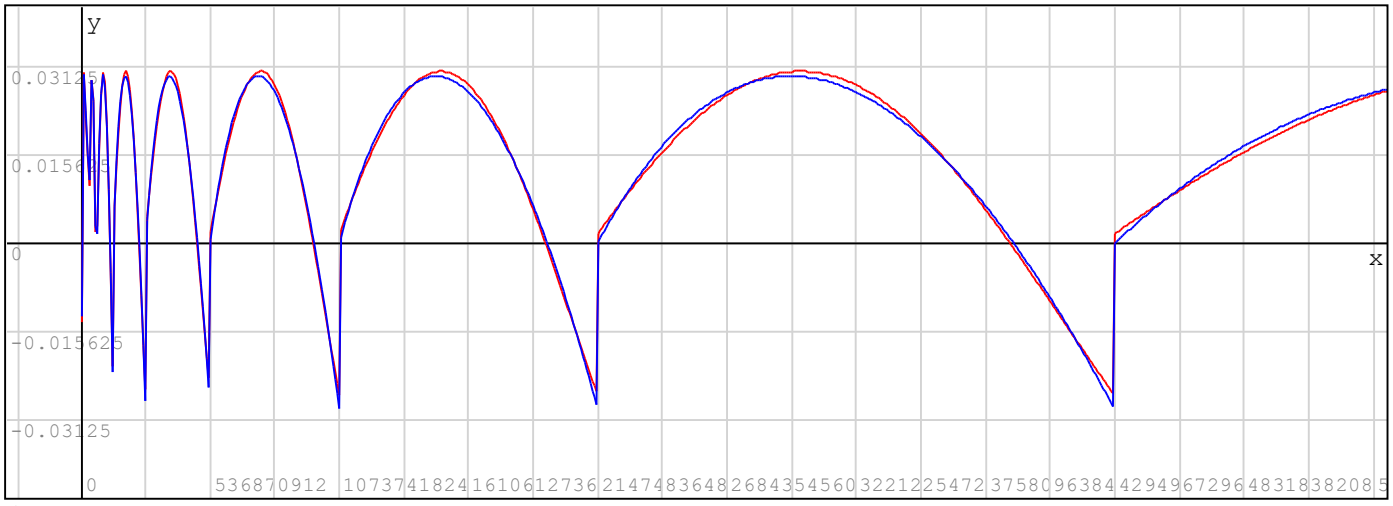
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$$linSqrtErr(x) := \sqrt{x} - linSqrt(x)$$

$$cosShift := \frac{\sqrt{2}}{5} + 0.01 \qquad cosAngle := 19.4 \qquad cosScale := 0.83$$

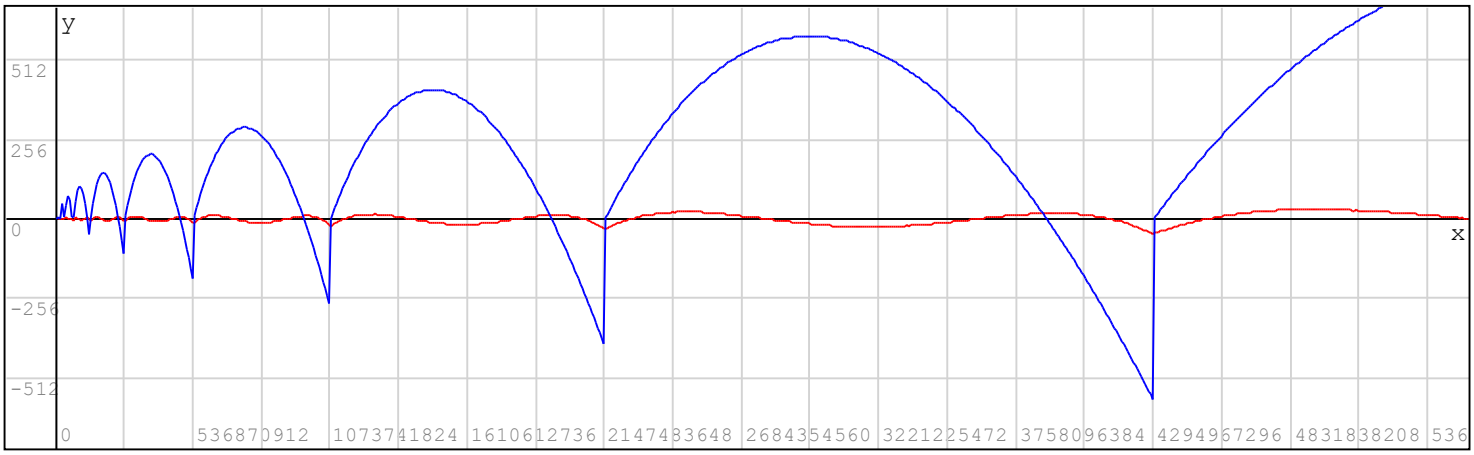
$$ec(x) := \left(\frac{\text{trunc}\left(\cos\left(\frac{x - \left(2^{msb(x)-1} + 2^{msb(x)-2} \right)}{2^{msb(x)}} \cdot 2 \cdot \pi + \pi \cdot \frac{cosAngle}{180} \right) \cdot \left(2^{32} - 1 \right) \right)}{2^{32}} - cosShift}{linScale(x)} \right) \cdot \frac{\sqrt{2}^{msb(x)}}{2^{6}} \cdot cos$$



$$\begin{cases} \dfrac{linSqrtErr(x)}{linScale(x)} \\ ec(x) \end{cases}$$

$$cosSqrt(x) := linSqrt(x) + (ec(x) \cdot linScale(x))$$

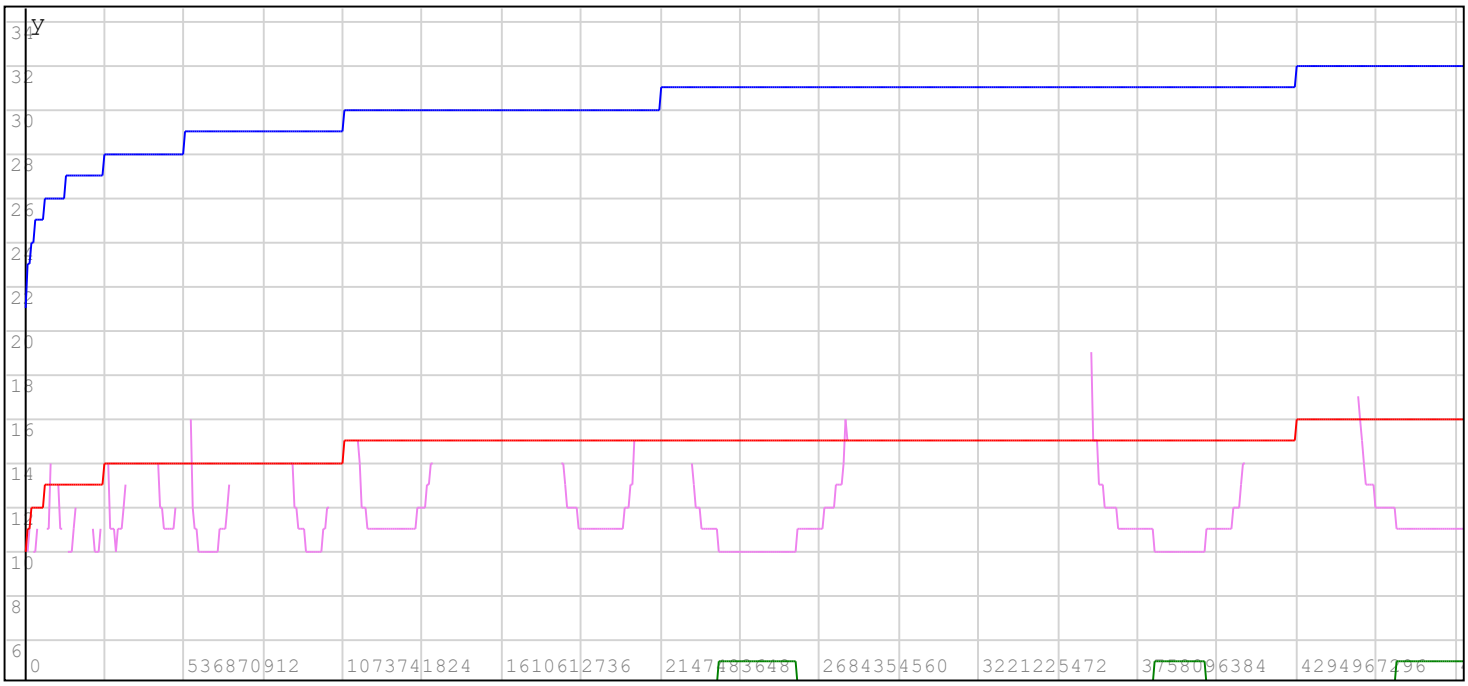
$$cosBitError(x) := \text{trunc}\left(\log_2\left(\left| \sqrt{\text{trunc}(x)} - cosSqrt(x) \right| \right) + \frac{2^{32}-1}{2^{32}} \right)$$



$$\begin{cases} \sqrt{x} - linSqrt(x) \\ \sqrt{x} - cosSqrt(x) \end{cases}$$

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$$\begin{cases} argBitCount(x) \\ resultBitCount(x) \\ cosBitError(x) \\ resultBitCount(x) - cosBitError(x) \end{cases}$$

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